International Journal of Nutrition and Food Sciences
2026, Vol. 15, No. 4, pp. 171–177
https://doi.org/10.11648/j.ijnfs.20261504.13

Review Article

The Science of Synbiotic GOS and L. Reuteri in Digestive Comfort, Reducing Colic-Related Symptoms, Regurgitation, and Advancing Muscle and Bone Development

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Abstract

Synbiotic nutrition, which combines prebiotics and probiotics, has gained increasing attention for its potential role in supporting digestive comfort, gut health, and healthy development during infancy and early childhood. The gastrointestinal tract of infants is still developing, and the establishment of a balanced intestinal microbiota is an important part of this process. During this period, digestive symptoms such as colic, excessive crying, fussiness, gas-related discomfort, irregular stool patterns, and regurgitation are common concerns for parents and caregivers. Nutritional strategies that support a healthy gut environment may therefore provide meaningful benefits. Galacto-oligosaccharides (GOS) are well-recognized prebiotics that selectively stimulate the growth and activity of beneficial intestinal bacteria, particularly bifidobacteria. By promoting a microbiota composition associated with a healthier intestinal environment, GOS may contribute to improved stool consistency, regularity, and overall digestive tolerance. A bifidobacteria-rich microbiota may also support the maturation of intestinal function and help maintain an appropriate balance between beneficial and potentially harmful microorganisms. Limos lactobacillus reuteri is a probiotic species that has been widely studied in relation to gastrointestinal health. It may support gut microbial balance, intestinal barrier function, and digestive comfort. Evidence suggests that L. reuteri may help improve gastrointestinal motility and reduce symptoms associated with infant colic, including excessive crying and fussiness. It may also play a role in reducing gas-related discomfort and regurgitation by supporting a more stable intestinal environment and better digestive function. When combined, GOS and L. reuteri may provide complementary or synergistic benefits. GOS can create a favorable environment for beneficial bacteria, while L. reuteri may contribute directly to microbial balance and gastrointestinal function. Together, they may support stool regularity, reduce digestive discomfort, and improve tolerance during early childhood. This combined approach may be particularly relevant during periods when the digestive system is still immature. Beyond digestive comfort, improved gut health may support nutrient digestion and absorption. Better utilization of protein, calcium, and other minerals is important for muscle strength, bone development, and physical growth. Adequate nutrient absorption during infancy and early childhood contributes to healthy musculoskeletal development and overall well-being. This review discusses the scientific basis for the combined use of GOS and L. reuteri in early childhood nutrition, with emphasis on digestive comfort, colic-related symptoms, regurgitation, nutrient utilization, and musculoskeletal development.

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Received: 18 May 2026; **Accepted:** 1 June 2026; **Published:** 27 July 2026

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